Quantifying the Court: Modeling cross-court groundstroke accuracy using weighted target zones and machine learning in collegiate men's tennis

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Abstract

Traditional tennis performance metrics classify shots as simply "in" or "out", obscuring meaningful differences in shot precision. This study developed a weighted target-zone scoring system to quantify cross-court groundstroke accuracy among five collegiate and recreational tennis players (1,072 shots total) by combining automated SwingVision tracking data with manually assigned accuracy zones. Statistical tests (t-tests, one-way and two-way ANOVA) and five machine learning regression models were used to evaluate whether stroke type, accumulated fatigue, player skill (UTR), and handedness predict shot accuracy. Individually, stroke type and fatigue stage showed statistically significant but small effects on accuracy. However, a two-way ANOVA revealed these two predictors are severely confounded ($r = -0.939$) due to a blocked data collection protocol, and neither retained significance once evaluated jointly. Across all five machine learning models, evaluated using leave-one-session-out cross-validation, none achieved positive predictive R^2, though several outperformed a naive baseline. Player skill (UTR) and shot direction emerged as the strongest correlates of accuracy in exploratory analysis. Substantial discrepancies were also identified between automated tracking outputs via SwingVision software and manually reviewed accuracy zones. Within this dataset and experimental design, these findings suggest that the tested predictors do not support reliable prediction of individual shot accuracy and that experimental design choices limit what can be concluded about fatigue's effect on groundstroke accuracy.

1. Background

Tennis performance depends on a player's ability to consistently place shots in strategically advantageous locations while responding to balls of varying speed, spin, and trajectory. At the collegiate level, players must continually adapt their shot selection and execution throughout a match, often with limited opportunities for individualized coaching. With limited coaching resources available for many players, most are required to take greater responsibility for evaluating their own performance and identifying opportunities for improvement. Advances in computer vision and player-tracking technologies have made it possible to collect detailed shot information during practice and competition. Systems such as SwingVision automatically estimate bounce locations, classify shot outcomes, and measure ball speed using smartphone video; providing athletes and coaches with considerably more information than traditional match statistics alone. This can greatly help players evaluate their performance and make identifying opportunities for required change much easier. Previous work by Whiteside and Reid (2018) demonstrated how structured observational methods can improve the classification of tennis shots, highlighting the value of combining automated tracking with human interpretation.

Although modern tracking software provides substantial amounts of information, many performance evaluations continue to rely on simple statistics such as winning shots, unforced errors, or whether a shot landed inside or outside the court. While these metrics remain useful, they fail to capture differences in shot precision. A ball landing near the intended target provides substantially greater tactical value for a player than one that barely lands inside the baseline, despite both being classified simply as "in." To address this limitation, this project developed a weighted accuracy scoring system that measures how closely each cross-court groundstroke landed in regard to an intended target location. By combining manually reviewed accuracy scores with automated tracking data collected through SwingVision, this study investigates how player characteristics, incoming ball conditions, stroke mechanics, and accumulated fatigue influence shot placement, leading to final accuracy determinations. The project demonstrates how data engineering, visualization, statistical analysis, and machine learning can be integrated to provide a complete sports analytics workflow that supports evidence-based coaching and athlete development.

2. Motivation

This study builds upon an earlier project completed in DATA 152, where a set of 48 shots was evaluated to examine the potential relationship between incoming ball speed and accuracy. The project subsequently expanded in both scope and complexity. The original capstone plan included a broader set of incoming-ball characteristics, but limitations identified during data collection and validation led to a refinement of the final analytical scope. The final study focused on stroke type, accumulated fatigue, player skill, and handedness as predictors of accuracy, while additional shot characteristics were examined exploratorily. Rather than treating these changes as a failure to follow the original plan, the revised scope reflects an iterative data science process in which the research question was refined to match the quality and structure of the collected data.

Unlike many sports analytics studies that rely exclusively on publicly available datasets, this project required the collection and validation of original experimental data. Every recorded shot was manually reviewed and assigned an accuracy zone, providing a level of detail that would not have been possible using automated tracking software alone. This process provided practical experience in experimental design, data engineering, feature engineering, statistical modeling, and machine learning while producing findings that may assist collegiate tennis players and coaches in understanding factors that influence shot accuracy.

3. Problem and Research Question

Collegiate tennis players frequently make decisions regarding technique, strategy, and practice focus using subjective observations rather than objective performance data. Although modern tracking systems provide extensive shot-level information, automated classifications and court coordinates are not always sufficiently accurate for detailed performance analysis. Without reliable performance measurements, players and coaches may overlook important factors that contribute to successful shot placement. Furthermore, relying solely on automated tracking outputs may introduce measurement error that limits the validity of statistical analyses and predictive models. This project aimed to develop a reproducible workflow that combines automated SwingVision data with manually validated accuracy scores. Rather than treating shot outcomes as a simple binary in-or-out variable, a weighted target-zone scoring system was developed to quantify shot precision. Statistical analysis and machine learning techniques were then applied to identify the variables most strongly associated with accurate cross-court groundstroke placement. The central research question guiding this study is:

To what extent do stroke type, accumulated fatigue, player skill, and handedness predict cross-court groundstroke accuracy among collegiate and recreational tennis players when accuracy is measured using a weighted target-based scoring system?

4. Impact and Stakeholders

The primary stakeholders for this project are the members of the Willamette University men's tennis team and the coaching staff. By providing a more detailed measure of shot precision than traditional in-or-out statistics, the project offers information that can help identify performance trends, evaluate practice effectiveness, and guide future player development. Beyond the immediate stakeholders, the methodology developed in this study demonstrates a reproducible approach for integrating automated player-tracking technologies with manual validation. Although the present study focuses on a single collegiate tennis program, the workflow can be adapted for larger datasets, different competitive levels, or alternative sports analytics applications.

5. Related Work

The rapid growth of sports analytics has transformed the ways in which athletic performance is measured and evaluated. Advances in computer vision and automated tracking technologies have enabled researchers and coaches to collect detailed shot-level information that was previously unavailable without extensive manual observation. Whiteside and Reid (2018) proposed a structured shot taxonomy for professional tennis that demonstrates the importance of observational classification when interpreting player performance. Their work influenced the methodological approach used in this study by supporting the integration of manual review alongside automated tracking outputs. Research has also demonstrated that fatigue influences athletic performance throughout extended periods of play. Fiorilli *et al*. (2020) observed measurable declines in physical performance following competitive tennis matches, supporting the use of elapsed session time as a practical proxy for accumulated fatigue when physiological measurements are unavailable. Consequently, elapsed session time was incorporated into the present analysis as one of the predictor variables. Recent developments in machine learning have further expanded the capabilities of tennis analytics. Lei *et al.* (2024) demonstrated that predictive algorithms can capture complex nonlinear relationships within tennis performance data that may not be identified through traditional statistical techniques alone. Their work motivated the inclusion of multiple predictive modeling approaches within this project to compare model performance and identify influential predictors of shot accuracy. Finally, O'Donoghue (2001) emphasized that tactical shot quality often provides more meaningful insight than simple match statistics. This concept directly motivated the development of the weighted target-zone scoring system used throughout the present study, which measures shot precision rather than relying solely on binary in-or-out classifications.

Although previous studies have examined fatigue, machine learning, and tactical shot analysis independently, relatively little research has combined manually validated shot placement with automated player-tracking data within a reproducible data science workflow. The present study seeks to address that gap.

6. Experimental Design

Participants. Data were collected from seven participants representing a range of tennis experience and skill levels. Participants included members of the Willamette University men's tennis team, the head men's tennis coach, and one recreational player. This convenience sample provided variation in playing ability while remaining feasible within the project's available data collection period. All participants provided informed consent before participating. Personal identifiers were removed during data processing to preserve participant anonymity. Five completed sessions were available for analysis at the time of this report, yielding five sessions (one per player) and 1,072 usable shots after data cleaning. Universal Tennis Rating (UTR) across these five participants ranged from 1.5 to 12.0, providing meaningful variation in skill level despite the reduced sample.

Experimental Procedure. Data collection was conducted using structured cross-court groundstroke drills designed to minimize tactical variation while producing a large number of comparable observations. To minimize variations in incoming ball characteristics, a ball machine was used to feed each ball. Participants repeatedly attempted to hit cross-court forehands and backhands toward a predefined target area while SwingVision recorded each session. The application automatically generated measurements including incoming ball speed, outgoing ball speed, estimated bounce coordinates, stroke type, spin classification, shot direction, and temporal information. Each recorded session was subsequently reviewed manually to assign weighted accuracy zones based on observed landing locations. This two-stage data collection process produced both automated tracking variables and manually validated performance measures for every shot. Each session followed a blocked protocol: participants completed approximately 100 forehand groundstrokes before transitioning to backhand groundstrokes, rather than alternating stroke type on a shot-by-shot basis. This design choice, made to simplify data collection logistics, has downstream implications for interpreting the fatigue and stroke-type results.

Weighted Accuracy Zones. A central contribution of this project is the development of a weighted target-zone scoring system. Rather than treating all successful shots equally, the intended target area was divided into sixteen manually defined accuracy zones. Each zone received a score based on its proximity to the ideal landing location. Shots landing closest to the target received the highest score (0.95), while progressively lower scores were assigned to surrounding regions. Any shot failing to land within a target zone received a score of 0.00. This scoring system provides substantially greater resolution than traditional binary outcome variables by distinguishing highly accurate shots from those that merely remain in play.

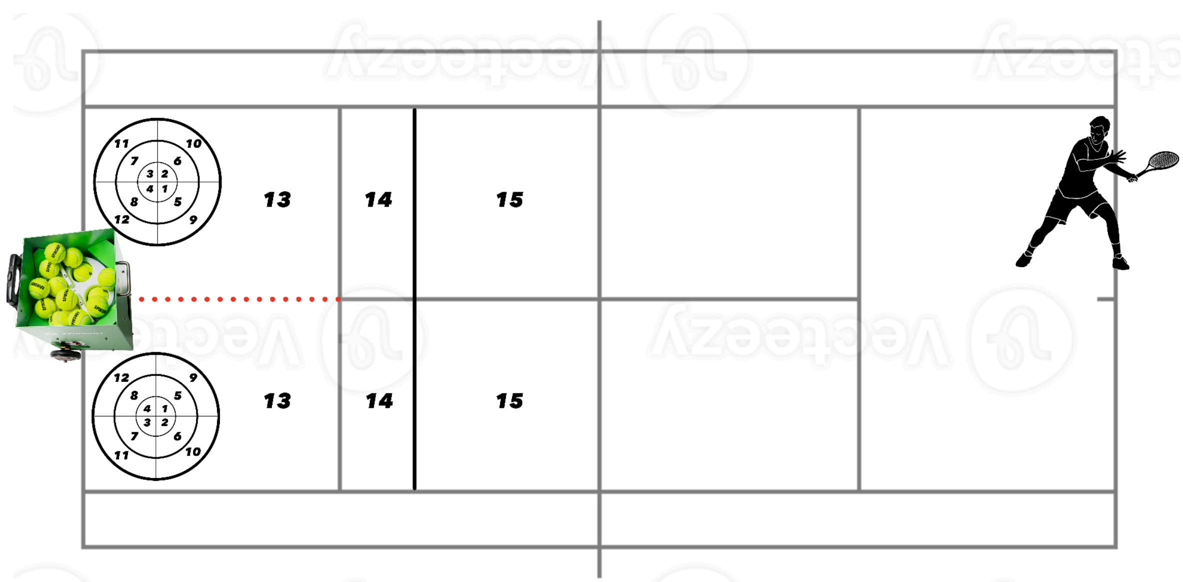

Figure 1. Weighted target-zone scoring system used to quantify cross-court groundstroke accuracy. Sixteen manually defined zones were assigned scores from 0.00 to 0.95 based on distance from the intended target, with shots outside the scoring zones receiving a score of 0.00. Weights were determined based off of crosscourt rally ball drills used in Willamette Men's Tennis practices.

7. Data and Data Engineering

Raw Data. The raw dataset consisted of a CSV export from SwingVision containing 1,076 shots across five sessions, with 24 recorded fields per shot including player identifiers, UTR, handedness, stroke type, spin classification, ball speed, bounce coordinates, hit coordinates, shot direction, shot outcome, timing information, and the manually assigned accuracy zone.

Cleaning and Validation. Five data quality issues were identified and addressed during cleaning: (1) One UTR value was recorded with a stray tilde character ("~12") and was corrected to a numeric value; (2) Ball speed values were checked against a plausible range for groundstrokes (5-100 mph), no values fell outside this range after correction of the mislabeled columns; (3) Elapsed session time was verified to be non-decreasing within each session, consistent with its derivation from continuous video timestamps; (4) Four shots were classified as volleys rather than forehand or backhand groundstrokes, these were retained in the cleaned dataset for transparency but excluded from the modeling dataset as the research question is specific to groundstrokes; (5) Player identifiers were anonymized to a standardized PLAYER_## format at the point of ingestion.

One additional coordinate anomaly was identified during exploratory visualization: a single shot recorded a bounce location approximately 45.7 feet from the net, more than six feet beyond the physical baseline (39 feet) and therefore impossible under real court geometry. Cross-referencing this shot's other recorded attributes (a legitimately scored accuracy zone alongside an "Out" outcome and a "deep" bounce classification past the baseline) suggested a probable tracking failure on a high-arcing ball rather than a data entry error. This coordinate was corrected using the median position of other shots sharing the same accuracy zone and stroke type, and is discussed further in Section 12.

Feature Engineering. The manually recorded accuracy zone (values 0-15) was converted into a continuous weighted accuracy score ranging from 0.00 (zone 0, a miss) to 0.95 (zone 1, the shot pattern closest to the target), using a predetermined mapping in which each zone was assigned a fixed score based on the target-zone design. Accumulated fatigue was operationalized in two complementary ways: as a continuous, session-normalized proportion of elapsed time (fatigue_pct, ranging from 0 at the start of a session to 1 at its end, allowing comparison across sessions of different total lengths), and as a categorical variable (fatigue_stage: early, mid, late) dividing each session into equal thirds by elapsed time. Stroke type and handedness were encoded as binary indicators for use in statistical and machine learning models. An initial implementation error was identified and corrected during development: the handedness indicator was originally coded to check for the string "right," but the underlying data used single-letter codes ("r"/"l"), causing the indicator to be incorrectly coded as zero for all shots regardless of true handedness. This was discovered through independent verification of the encoding logic before any results were generated from it. The final analysis dataset, after excluding volleys, contained 1,072 shots across five sessions, with 538 forehand and 534 backhand shots.

Repository Structure and Pipeline. Data processing was implemented as a single reproducible R script (`tennis_pipeline.R`) organized into three stages: raw data cleaning and validation, feature engineering, and exploratory visualization. Cleaned intermediate data and analysis-ready data were written to separate directories (`data/interim/`, `data/processed/`) to preserve traceability from raw input to final analytical dataset. Formal statistical testing was implemented separately in R (`statistical_tests.R`), and machine learning modeling was implemented in Python (`model.py`), reflecting the different tooling strengths of each language for these tasks.

8. Ethics and Responsible Use

All participants provided informed, written consent prior to their hitting sessions, in accordance with the project's IRB determination. Participants were informed that participation was voluntary and that opting out carried no social, athletic, or academic penalty. Because data collection involved individuals with existing personal and professional relationships to the researcher, including current teammates, the head coach, and a personal acquaintance; specific procedural safeguards were applied to preserve data integrity and protect participants from any real or perceived pressure to participate. The head coach's data was treated identically to all other participants' data and used only in aggregate as an anonymized "expert baseline" observation, without any indication in the analytical dataset of coaching authority or evaluative relationship to other participants. All participants' data, regardless of relationship to the researcher, were processed through the same automated cleaning and scoring pipeline with no manual overrides applied selectively to any individual's records. Participant names were never retained in any analytical file. Player identifiers were replaced with randomized alphanumeric codes (e.g., PLAYER_04) at the point of data ingestion, and UTR and handedness were retained only as anonymized numeric and categorical attributes disconnected from any identifying information. Because manual accuracy scoring required subjective judgment by a single reviewer, this study also carries a responsibility to disclose that limitation openly rather than presenting manually assigned scores as an objective ground truth.

9. Methods

Statistical Testing. To directly test the research question, three sets of formal statistical tests were conducted using accuracy_score as the outcome variable. First, an independent-samples t-test (Welch's variant, selected based on a significant Levene's test for unequal variances) compared mean accuracy between forehand and backhand shots. Second, a one-way analysis of variance (ANOVA) compared mean accuracy across the three fatigue stages (early, mid, late), with Tukey's HSD used for post-hoc pairwise comparisons. Third, and most directly relevant to the research question, a two-way ANOVA using Type II sums of squares evaluated stroke type, fatigue stage, and their interaction simultaneously, allowing each predictor's effect to be assessed while accounting for the other. Effect sizes (Cohen's d for the t-test; eta-squared for the ANOVA models) were reported alongside p-values throughout, given the large sample size ($n = 1,072$), which provides substantial statistical power to detect even practically negligible effects.

Machine Learning. Five regression approaches were compared to evaluate whether a broader set of predictors, stroke type, fatigue (as a continuous proportion), a stroke type by fatigue interaction term, handedness, and UTR; could predict individual shot accuracy: Linear Regression, K-Nearest Neighbors, Support Vector Regression (radial basis function kernel), Random Forest, and Gradient Boosting. Rather than selecting a single algorithm in advance, all five were evaluated under identical conditions and the best-performing model was reported as the result, consistent with recommendations from Lei *et al.* (2024) regarding the value of comparing multiple algorithmic approaches. Model evaluation used leave-one-session-out cross-validation (five folds, one per player/session) rather than a random shot-level split. Because individual shots within a session are repeated measures from the same player rather than independent observations, a random split would allow a model to learn player-specific patterns during training and then be evaluated on further shots from that same player, artificially inflating apparent performance. Leave-one-session-out cross-validation instead evaluates each model's ability to predict an entirely unseen player's shots, which is a substantially more honest test of generalization given the study's goals. Model performance was assessed using mean absolute error (MAE), root mean squared error (RMSE), and R^2, compared against a naive baseline model that always predicts the training set's mean accuracy score. Following the success criterion established in the original project proposal, a model was considered successful if it achieved R^2 greater than 0.0 under this cross-validation scheme. Hyperparameters for each tunable model were subsequently optimized via grid search, using the identical cross-validation folds to ensure tuning decisions were evaluated under the same honest, session-held-out standard rather than a separate, potentially leaky validation scheme.

Exploratory and Correlational Analysis. In addition to the formal RQ1 tests, Pearson correlation coefficients were computed between accuracy_score and all available numeric and dummy-coded categorical predictors, including variables outside the original RQ1 scope (shot direction, spin type, ball speed, and bounce location), to identify any stronger predictors of accuracy that the primary analysis might have overlooked.

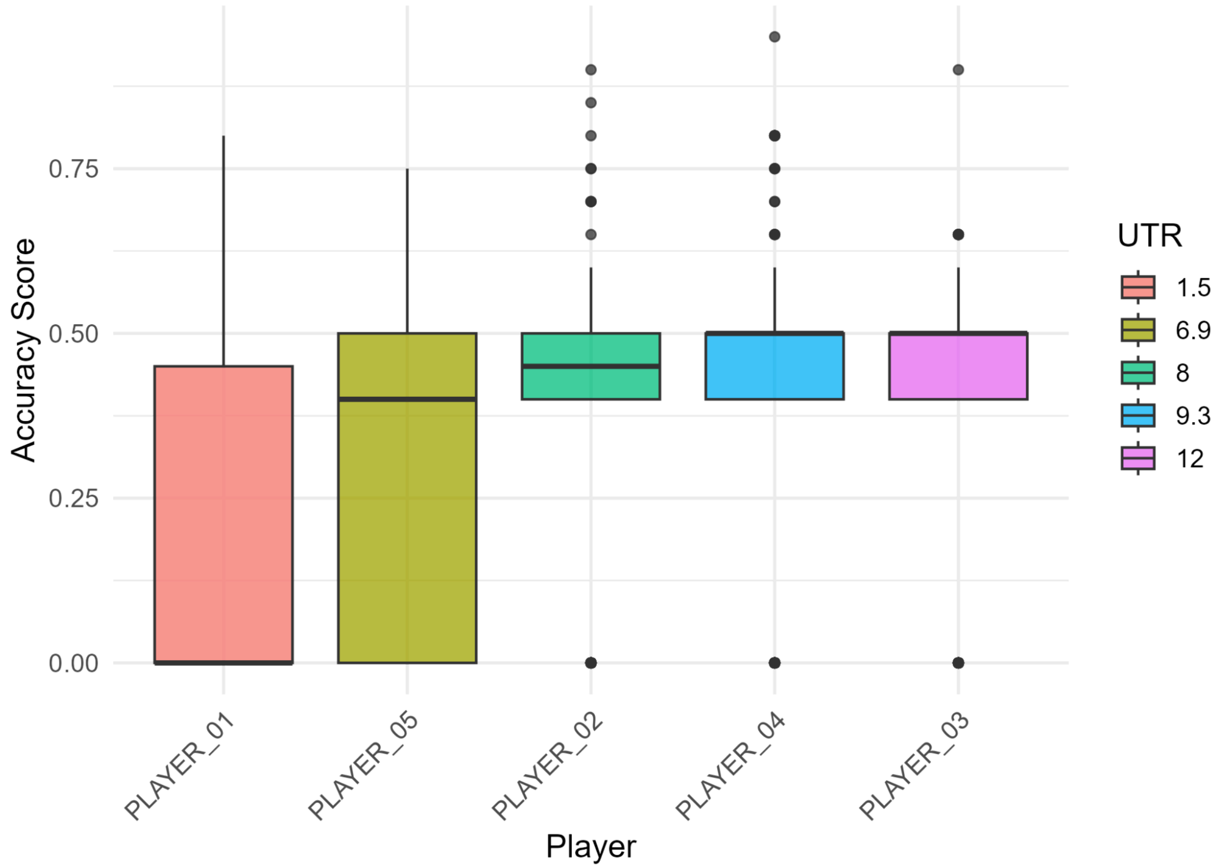

Figure 2. Distribution of weighted accuracy scores by player and ordered by Universal Tennis Rating (UTR). Boxplots illustrate differences in shot accuracy across participants with varying levels of player skill.

10. Results

Overview. Of 1,072 analyzed shots, 328 (30.6%) received an accuracy zone of 0 (a miss), while 85 shots (7.9%) landed within one of the twelve innermost scored target rings, and 659 shots (61.5%) landed within the three progressively larger surrounding scoring regions. Mean accuracy score was 0.361 for forehand shots (SD = 0.229) and 0.318 for backhand shots (SD = 0.239).

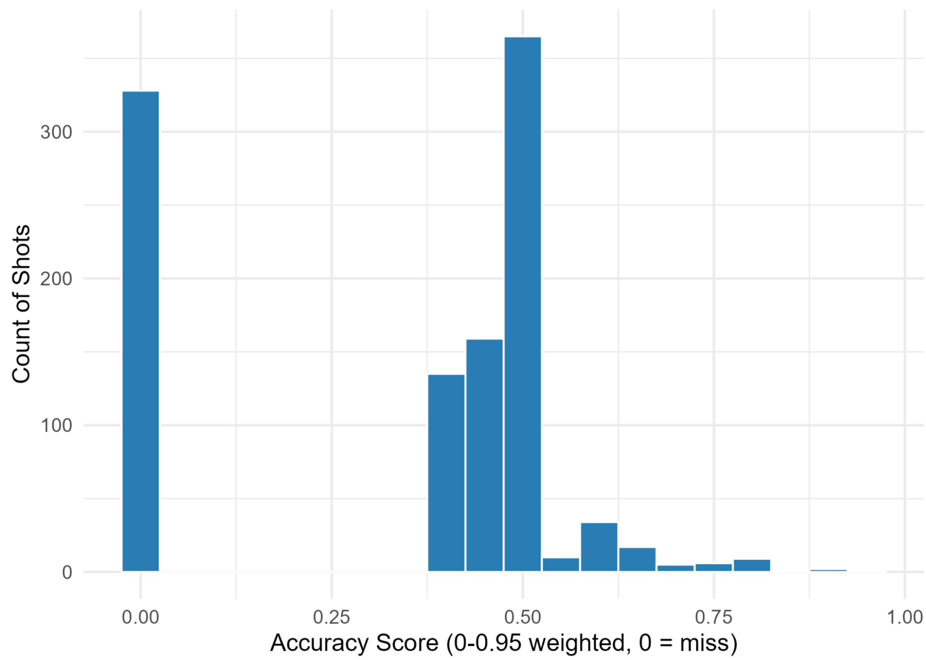

Figure 3. Distribution of weighted accuracy scores across 1,072 analyzed groundstrokes. The distribution shows the concentration of shots at lower and intermediate accuracy scores, with shots receiving a score of 0.00 representing misses outside the defined scoring zones.

RQ1: Statistical Tests. Tested independently, both stroke type and fatigue stage showed statistically significant effects on accuracy score. An independent t-test found forehand accuracy significantly higher than backhand accuracy (Welch's $t = 3.04$, $p = .002$), though the effect size was small (Cohen's $d = 0.19$). A one-way ANOVA found a significant effect of fatigue stage ($F = 4.61$, $p = .010$), with accuracy declining modestly from the early stage ($M = 0.361$) to the late stage ($M = 0.316$) of a session; this effect size was likewise small, with fatigue stage explaining under one percent of total variance in accuracy score (eta-squared = 0.0085). However, when both predictors were evaluated together in a two-way ANOVA, neither stroke type ($F = 0.33$, $p = .565$), fatigue stage ($F = 0.16$, $p = .853$), nor their interaction ($F = 0.93$, $p = .396$) reached statistical significance. This divergence from the individual tests was traced to a severe confound between the two predictors: because each session followed a blocked protocol (forehands completed before backhands), stroke type and normalized elapsed time correlate at $r = -0.939$ in this dataset. Nearly all forehand shots occurred in the early fatigue stage (441 of 538) and nearly all backhand shots occurred in the late stage (496 of 534), with essentially no overlap. As a result, the two-way model cannot attribute shared variance uniquely to either predictor. This finding, and its implications for interpreting RQ1, is discussed further in Discussions and Limitations.

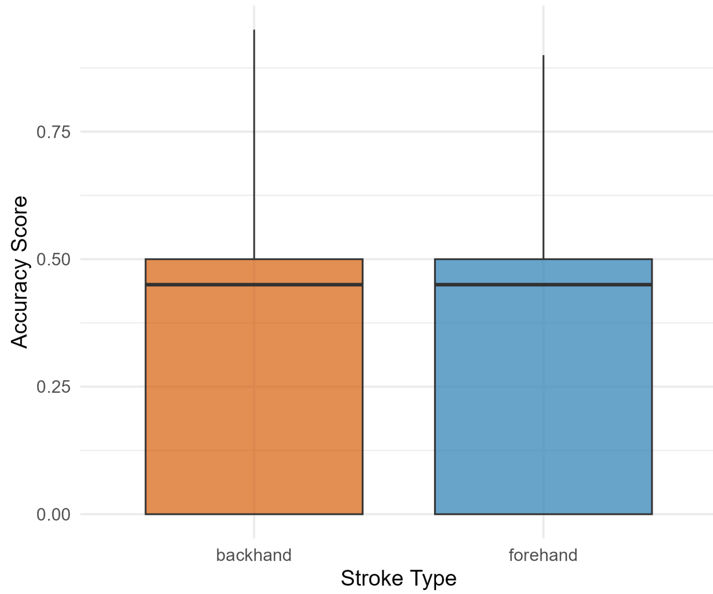

Figure 4. Weighted accuracy scores by stroke type. Boxplots compare the distribution of accuracy scores for forehand and backhand groundstrokes. Forehands had a higher mean accuracy score than backhands (0.361 vs. 0.318), although the observed effect was small.

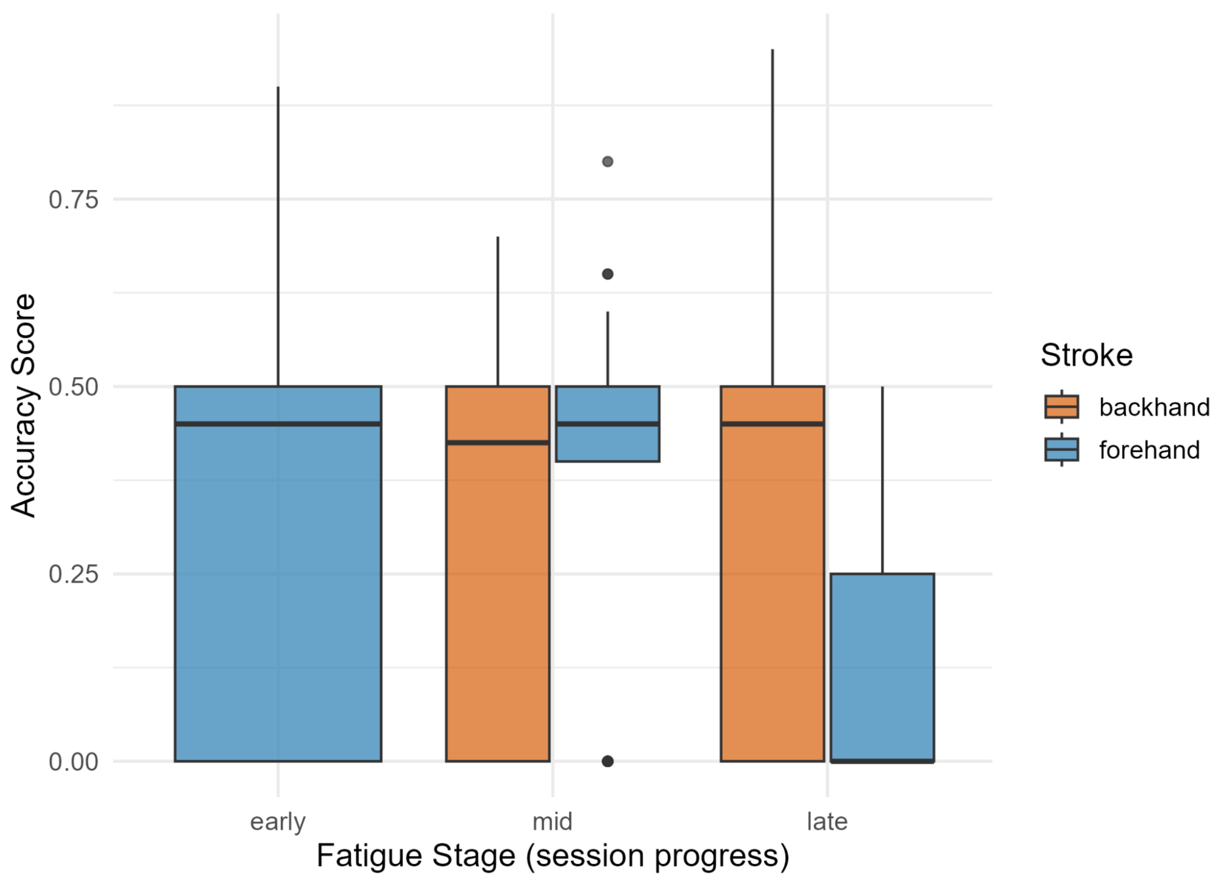

Figure 5. Weighted accuracy scores across fatigue stages, separated by stroke type. Boxplots show the distribution of accuracy scores during the early, middle, and late portions of each session for forehand and backhand groundstrokes.

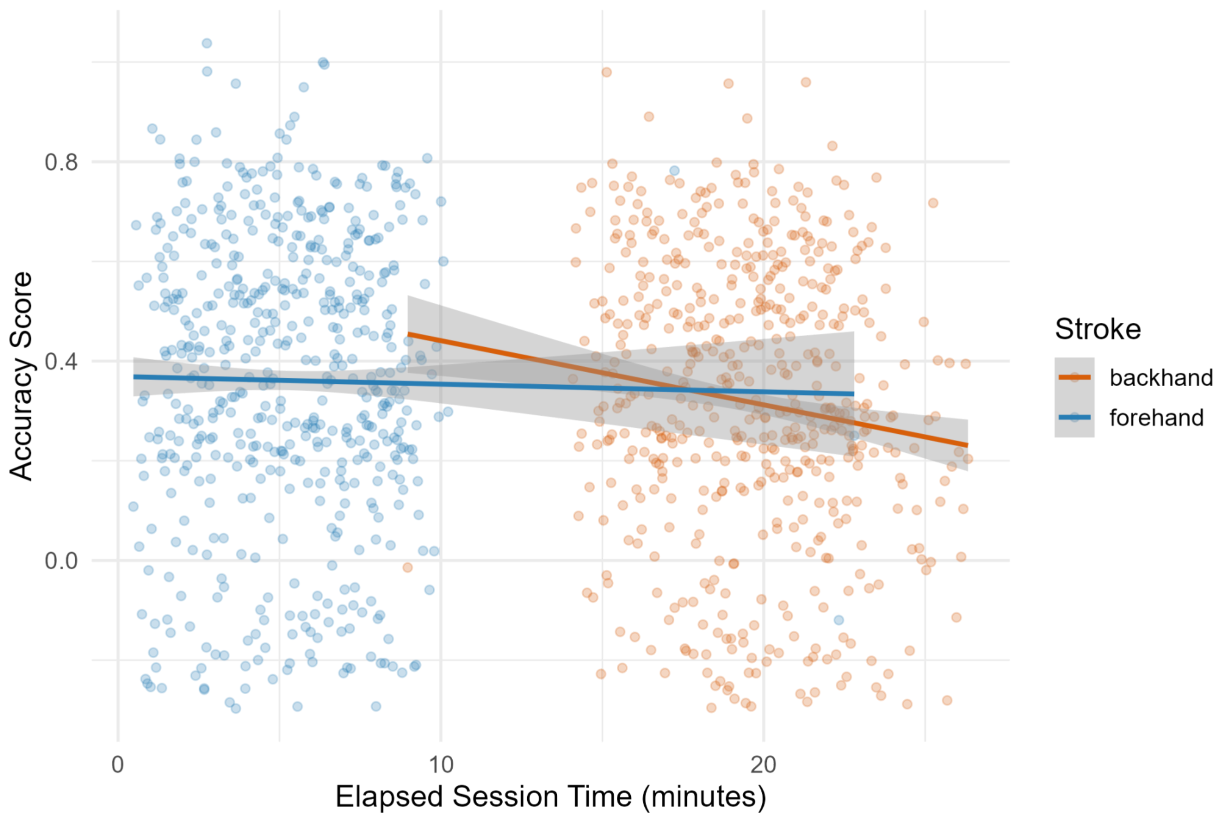

Figure 6. Relationship between weighted accuracy score and total elapsed session time. Individual shots are shown by stroke type with fitted trend lines and confidence bands. The figure illustrates the observed relationship between accumulated fatigue, represented by total elapsed session time, and accuracy while highlighting the substantial confounding between fatigue and stroke type resulting from the blocked collection protocol.

Machine Learning Model Comparison.

Table 1. Cross-validated performance for the five evaluated regression models and the naive mean baseline, before and after hyperparameter tuning.

Model	Default R^2	Tuned R^2	Tuned MAE
Random Forest	-0.221	**-0.190**	0.187

SVR (RBF)	-0.384	-0.196	0.174
Gradient Boosting	-0.299	-0.214	0.192
Linear Regression	-0.250	-0.250 (no tuning applicable)	0.194
KNN	-0.319	-0.280	0.185
Baseline (mean)	-0.304	—	0.217

Note: XGBoost was included in the original modeling plan but was unavailable in the environment used to generate these results and was therefore not evaluated.

No model achieved positive R^2 under leave-one-session-out cross-validation, meaning the original proposal's success criterion ($R^2 > 0.0$) was not met by any model tested. However, all five tuned models outperformed the naive baseline (R^2 ranging from -0.190 to -0.280, versus -0.304 for the baseline), indicating some recoverable signal even though absolute predictive accuracy for an unseen player remained limited. Tuning improved every tunable model's performance relative to its default configuration, most notably for SVR (R^2 improved from -0.384 to -0.196). The best-performing configuration, a tuned Random Forest with a maximum tree depth of three and a minimum of ten samples per leaf, favored a comparatively simple, constrained model over a more complex one, consistent with a dataset offering limited signal relative to its noise. Feature importance analysis across tree-based models consistently ranked UTR as the strongest predictor (66-72% relative importance), followed by the stroke-type-by-fatigue interaction term (12-14%) and fatigue_pct alone (11-18%); stroke type and handedness alone contributed minimally. Linear regression coefficients told a broadly consistent story: the stroke-type-by-fatigue interaction carried the largest coefficient (-0.19), followed by stroke type alone (+0.14) and fatigue_pct alone (+0.09), with UTR and handedness contributing comparatively little in this linear specification.

Exploratory Correlations. Beyond the core RQ1 predictors, Pearson correlations with accuracy_score identified shot direction as the single strongest correlate found in this study: shots recorded as traveling cross-court (matching the intended target direction) correlated with accuracy at $r = 0.486$, exceeding UTR ($r = 0.384$) and all other variables tested, including ball speed ($r = 0.197$) and spin type ($|r| < 0.06$ for all spin categories). This finding is intuitive given that the scoring target was specifically positioned for cross-court shots.

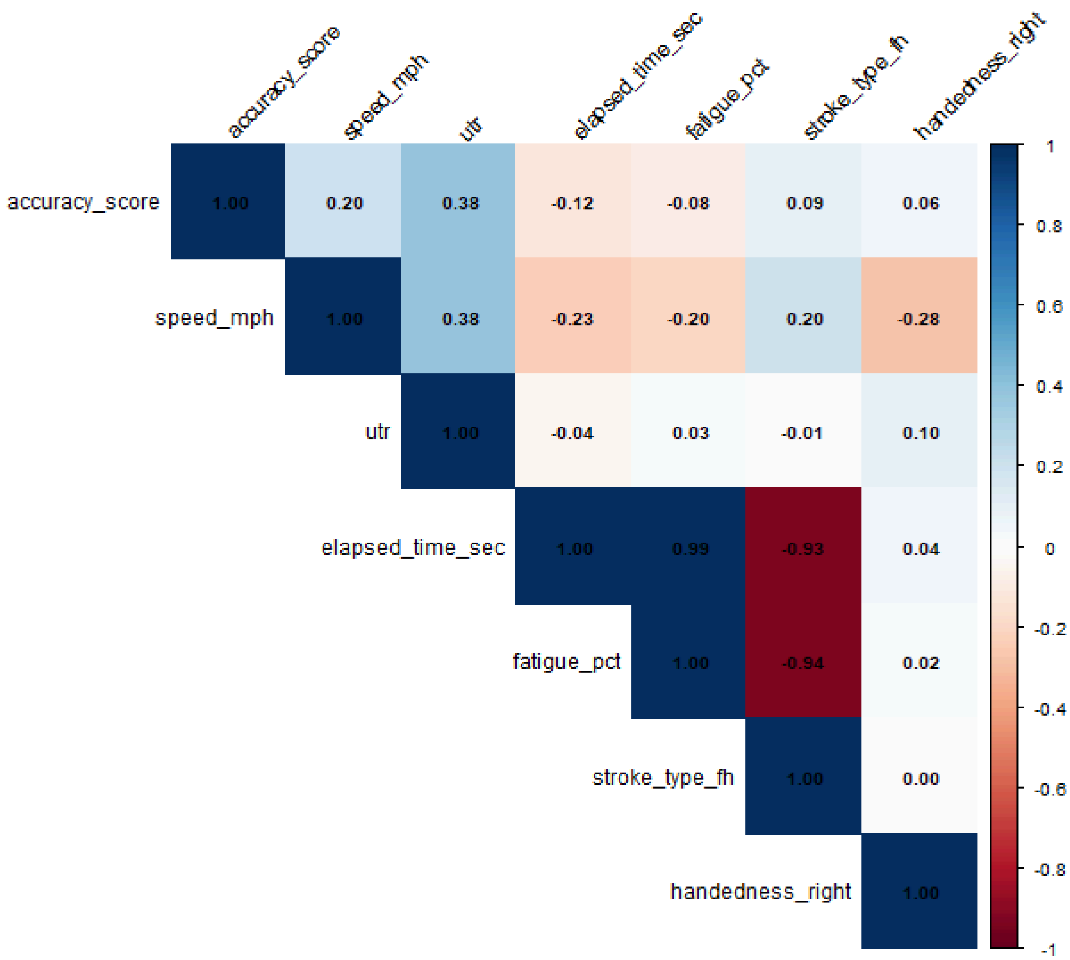

Figure 7. Correlation matrix of weighted accuracy score and predictor variables. Pearson correlation coefficients summarize the linear relationships among accuracy, player characteristics, fatigue, stroke characteristics, and other measured shot variables.

Spatial Findings. Reconstruction of the physical target geometry, based on the target's documented offset from the singles sideline and baseline, revealed that actual shot placement fell substantially short of the intended target across all participants. The highest-rated participant (UTR 12.0) never recorded a bounce beyond 32.1 feet from the net across the entire session, despite the target's center lying approximately 36 feet from the net. Mean bounce depth across all five participants ranged from 17.3 to 21.7 feet, indicating shots typically landed in the forward portion of the court well short of the intended target region regardless of player skill level.

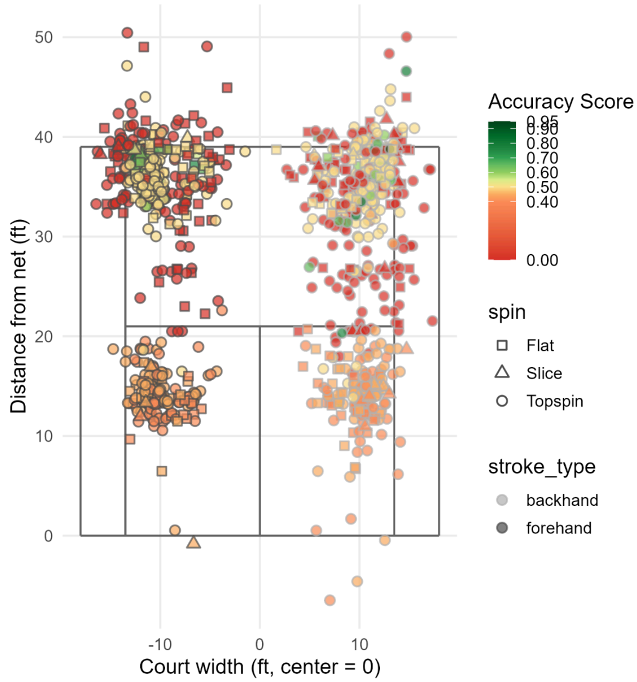

Figure 8. Spatial distribution of recorded shot bounce locations across 1,072 groundstrokes. Points are colored by weighted accuracy score and shaped by spin type. Bounce coordinates were obtained from SwingVision, with y-position adjustments applied to scored shots to better align recorded locations with the manually defined target-zone geometry.

11. Discussion

The central finding of this study is not a positive result in the traditional sense, but a methodologically important negative one: this dataset does not support a clean, independent estimate of either stroke type's or accumulated fatigue's effect on cross-court groundstroke accuracy, because the two predictors were not varied independently during data collection. Individually, each predictor showed a statistically detectable but practically small association with accuracy, consistent in direction with prior fatigue research (Fiorilli *et al.*, 2020) suggesting that performance modestly declines over the course of an extended physical session. However, the two-way analysis, the test that actually corresponds to the research question as originally framed, found no significant effect for either predictor once evaluated jointly, and machine learning models incorporating stroke type, fatigue, UTR, and handedness did not achieve positive predictive R^2 under honest cross-validation, although all five tuned models modestly outperformed the naive baseline. This result should not be interpreted as evidence that stroke type and fatigue have no relationship to shot accuracy. Rather, it reflects a specific and identifiable limitation of the blocked data collection protocol used in this study. The feature importance analysis consistently identified the fatigue-stroke interaction term as informative even where linear main effects were weak, suggesting a nonzero but modest and difficult-to-isolate relationship between these variables and accuracy.

Two findings emerged that were not part of the original RQ1 framing but merit discussion. First, UTR was consistently the strongest predictor identified across both the correlational analysis and the machine learning feature importance results, more informative than any variable directly tied to the stroke-fatigue research question. This suggests that overall player skill accounts for meaningfully more variance in shot accuracy than the specific mechanical and fatigue-related factors this study set out to isolate, a pattern broadly consistent with O'Donoghue's (2001) argument that tactical shot quality is shaped by a broad constellation of player factors rather than any single mechanical variable in isolation. Second, whether a shot traveled in the intended (cross-court) direction was the single strongest correlate of accuracy identified in this study. While partly a consequence of how the target itself was positioned, this finding underscores that shot accuracy, as measured here, is inseparable from shot selection and execution intent, a shot cannot be scored as accurately against a target it was never aimed at, regardless of any other characteristic. The consistent shortfall between actual shot depth and the target's intended location also warrants discussion. Every participant, regardless of skill level, landed shots well short of the target depth on average. This may reflect a genuinely difficult target design relative to what a groundstroke drill of this kind can realistically produce, and it meaningfully compresses the effective range of accuracy scores observed relative to the scoring system's full 0.00-0.95 scale. Future iterations of this scoring system may benefit from calibrating target difficulty against pilot data before full-scale collection.

12. Limitations

Several limitations affect the interpretation and generalizability of the findings presented in this study. These limitations span the experimental design, the reliability of automated tracking data, the manual accuracy scoring process, and the predictive performance of the models tested.

Sample Size and Statistical Power. Data collection ultimately included five of the seven participants originally planned, yielding five recorded sessions and 1,072 total shots. While this provided a reasonably large number of shot-level observations, the effective sample size for between-player comparisons remains small. Machine learning models were evaluated using leave-one-session-out cross-validation, in which each fold withholds an entire player's data rather than a random subset of shots; under this evaluation, no model, including tuned Random Forest, Gradient Boosting, SVR, KNN, and Linear Regression; achieved positive predictive R^2, indicating that none of the models tested could reliably predict a held-out player's accuracy from stroke type, fatigue, UTR, and handedness alone. This is consistent with substantial between-player heterogeneity that a sample of five players cannot adequately characterize, and it should be read as a property of the available data rather than as evidence against the modeling approaches themselves. Several models did outperform a naive mean-baseline under the same cross-validation scheme, suggesting some recoverable signal, but the absolute predictive power achieved should be interpreted cautiously given the sample size.

Confounded Experimental Design. A significant limitation emerged from the structure of the data collection protocol itself. Each session followed a blocked format in which participants completed approximately 100 forehand groundstrokes before transitioning to backhands, rather than alternating stroke type shot-by-shot. As a result, stroke type and accumulated session fatigue are severely confounded in the resulting dataset ($r = -0.939$ between stroke type and normalized fatigue). A two-way analysis of variance testing stroke type, fatigue stage, and their interaction simultaneously found none of the three terms statistically significant, despite each showing a significant effect when tested in isolation (independent t-test for stroke type, $p = .002$; one-way ANOVA for fatigue stage, $p = .010$). This divergence is a textbook multicollinearity artifact: because stroke type and fatigue stage co-occur almost deterministically within this dataset, neither variable's independent contribution to accuracy can be statistically isolated from the other. Consequently, this study cannot determine whether stroke type, accumulated fatigue, or some combination of the two drives the modest effects observed. Future data collection using an interleaved or randomized stroke-order protocol would be necessary to disentangle these predictors.

Automated Tracking Data Reliability. Several inconsistencies were identified in the automated SwingVision tracking output during data cleaning. The exported column headers did not consistently correspond to their underlying content and required correction based on inspection of the data itself rather than the file's own labels. Additionally, bounce coordinates disagreed with the corresponding manually assigned accuracy zone for a substantial share of scored shots, approximately 38 percent under a reconstructed target-zone geometry, and 117 shots (approximately 11 percent of the dataset) showed a direct contradiction between the automated in/net/out outcome field and the manually recorded accuracy zone. One shot recorded a bounce location approximately 45.7 feet from the net, six feet beyond the physical baseline, most likely reflecting a tracking failure on a high-arcing ball; this coordinate was manually corrected prior to analysis. Given these inconsistencies, the manually assigned accuracy zone was treated as the authoritative measure of shot accuracy throughout this study, while automated coordinates and outcome labels were used only for supplementary spatial visualization rather than for statistical modeling. This decision, while necessary, means that the spatial visualizations in this report are illustrative approximations rather than precise reconstructions of shot placement.

Manual Accuracy Zone Assessment. Although manually assigned accuracy zones were treated as more reliable than automated tracking outputs, this scoring process introduces its own limitations. All accuracy zones were assigned by a single reviewer without a formal inter-rater reliability check, leaving open the possibility of inconsistent judgment across sessions, particularly for shots landing near zone boundaries. Additionally, real shot placement fell substantially short of the intended target location across all players regardless of skill level: only 85 of 1,072 shots landed within the innermost scored target rings, and even the highest-rated participant (UTR 12.0) never recorded a shot within 32 feet of the intended target center, despite the target lying approximately 36 feet from the net. This consistent shortfall may reflect a target difficulty that exceeded what this drill format could realistically produce, and it substantially reduces the effective range of accuracy scores observed in the dataset relative to its theoretical 0.00-0.95 scale.

Predictive Model Performance. As noted above, no model tested; spanning linear, distance-based, and ensemble tree methods, achieved a cross-validated R^2 greater than zero for predicting individual shot accuracy from stroke type, fatigue, UTR, and handedness. While hyperparameter tuning improved several models' performance relative to their defaults, this improvement should be interpreted cautiously: with only five sessions available for tuning, hyperparameter selection is closer to identifying configurations that fit these five players well than to a validated demonstration of generalization to new players. The consistent underperformance across model types, rather than any single model's weakness, suggests that the available predictors do not capture enough of the variance in shot accuracy to support reliable individual-level prediction, and that additional variables, potentially including incoming ball characteristics, which were part of the original project scope but were not included in the final predictor set; may be necessary to improve predictive performance in future work.

13. Future Work

Several extensions to this study would directly address its most significant limitations. Most importantly, future data collection should use an interleaved or randomized stroke-order protocol, alternating forehand and backhand shots rather than completing one stroke type before the other, to break the severe confound between stroke type and accumulated fatigue identified in this study. This would allow each predictor's independent contribution to accuracy to be meaningfully estimated for the first time. Expanding the participant pool to the originally planned seven players, and ideally beyond, would improve the statistical power of between-player comparisons and provide more sessions for honest cross-validation of machine learning models, directly addressing the small-sample limitations. A formal inter-rater reliability check, in which a second reviewer independently scores a subset of shots against the same target-zone system, would strengthen confidence in the manually assigned accuracy scores that this study ultimately relied upon as ground truth. Incoming ball speed and other characteristics of the ball as it approached the player, part of the original project scope but excluded from the final analysis, represent a promising avenue for improving predictive performance, particularly given that none of the models tested in this study achieved positive predictive R^2. Additional candidate predictors identified through exploratory analysis, particularly shot direction and ball speed, showed stronger correlations with accuracy than the original RQ1 predictors and warrant formal inclusion in future predictive models. Given time constraints on this project, a mixed-effects model was not fit for the present analysis, though given the repeated-measures structure of this data (multiple shots nested within players), this approach with a random intercept per player may offer a more statistically appropriate alternative to the fixed-effects methods used in this study, particularly for future datasets with more participants. Although the present analysis treated accumulated fatigue as a continuous linear predictor, fatigue may influence performance nonlinearly across a session. Figure 9 presents a conceptual sigmoid relationship as a hypothesis for future investigation rather than an empirical fit to the current data. A larger dataset collected with an improved experimental protocol could test whether accuracy declines gradually, changes rapidly after a threshold, or follows another nonlinear pattern.

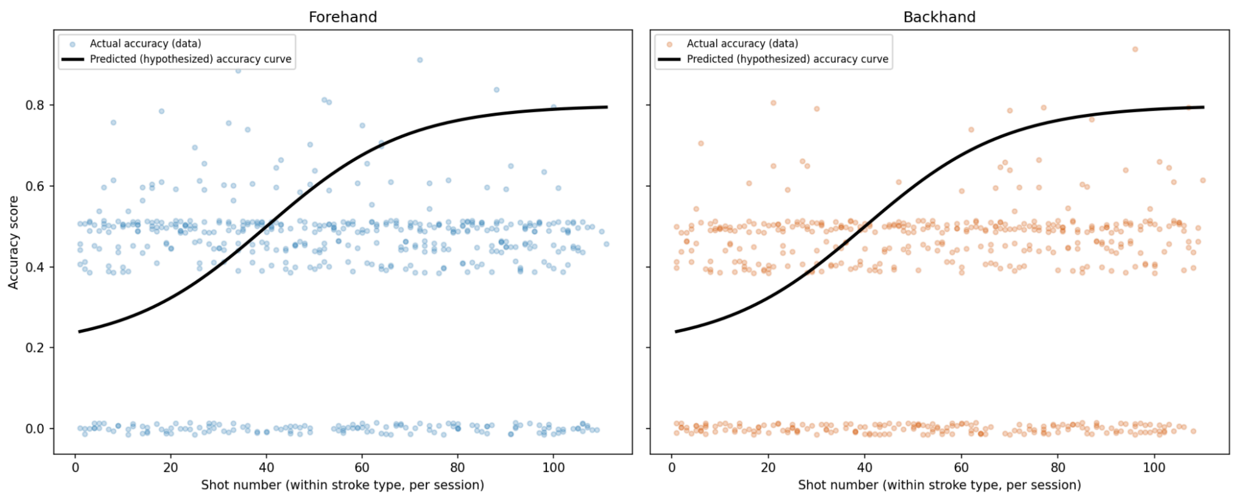

Figure 9. Conceptual hypothesis for a nonlinear relationship between accumulated fatigue and groundstroke accuracy. The sigmoid curve represents a proposed relationship in which accuracy changes at different rates across stages of accumulated fatigue. The curve is conceptual and was not fitted to the present dataset.

14. Conclusions

This study developed and applied a weighted target-zone scoring system to measure cross-court groundstroke accuracy with greater precision than traditional binary outcome statistics, and used this system to evaluate whether stroke type, accumulated fatigue, player skill, and handedness predict shot accuracy among collegiate and recreational tennis players. Rather than a straightforward confirmatory result, this study's central finding is that the data collection protocol used, a blocked practice format completing all forehands before all backhands, created a severe confound between stroke type and fatigue that prevents either predictor's independent effect from being statistically isolated. Machine learning models incorporating both predictors alongside player skill and handedness were unable to reliably predict individual shot accuracy for a held-out player under honest cross-validation, though several models modestly outperformed a naive baseline. These results do not indicate that stroke type and fatigue are irrelevant to shot accuracy; rather, they demonstrate the importance of experimental design in isolating causal or predictive relationships, and they identify player skill and shot selection (direction) as stronger correlates of accuracy than the specific mechanical factors this study set out to examine. Beyond its specific findings, this project's primary contribution is a reproducible, transparent workflow, spanning data engineering, validation, statistical testing, and machine learning; that can be directly extended with additional participants, an improved data collection protocol, and additional predictor variables in future work.

15. Reproducibility

All data processing, statistical analysis, and modeling code developed for this project is organized in a version-controlled repository structured as follows:

```None
data/raw/            original SwingVision export

data/interim/         cleaned data, post-validation

data/processed/        analysis-ready dataset used for all statistics and modeling

src/tennis_pipeline.R      data cleaning, feature engineering, and EDA (R)

src/statistical_tests.R    t-test, one-way and two-way ANOVA (R)

src/model.py                  machine learning model comparison and tuning (Python)

outputs/               all generated figures and result tables
```

The pipeline is designed to run end-to-end: `tennis_pipeline.R` produces the cleaned and analysis-ready datasets and all exploratory figures; `statistical_tests.R` and `model.py` can then be run independently against the processed dataset to reproduce the statistical and machine learning results. R scripts require the tidyverse, ggplot2, corrplot, and car packages; the Python modeling script requires pandas, numpy, scikit-learn, and matplotlib, with XGBoost as an optional additional model if installed. Raw participant data is not publicly distributed due to the scope of participant consent, but the anonymized processed dataset and all analysis code are available to reproduce every statistic, table, and figure reported in this document from that anonymized starting point. The GitHub repository for this project is available at: https://github.com/lcasano26/data510-Luca-Capstone.

16. References

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